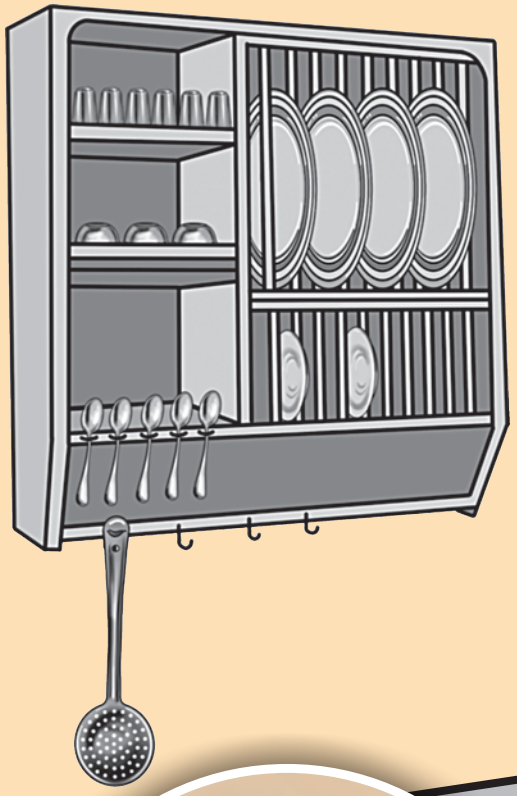
MATHEMATICS

STANDARD ONE



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गणित इयत्ता पहिली (इंग्रजी माध्यम)

- **Tele-MANAS Program**

Tele-MANAS (Tele Mental Health Assistance and Networking Across States) (14416/ 1800-89-14416)

Under the National Mental Health Program, the centrally sponsored scheme 'Tele-MANAS' is being implemented in the state of Maharashtra from October 2022. Under the Tele-MANAS Program, free 24 × 7 integrated and comprehensive mental health services are provided through the helpline number 14416. This includes treatment and counselling by experts for various mental problems. The information provided by you after calling the Tele-MANAS helpline remains completely confidential, so students, parents and teachers should feel free to call the helpline number and tell them about their problems and doubts and get guidance on it. Tell others to take help from this facility as well.

Tele-MANAS app has been launched by the Central Government and is also available in Marathi language. This app provides complete information about mental health. Tele-MANAS app is available free of cost.

This textbook aligns with the curriculum and syllabus developed by the State Council of Educational Research and Training (SCERT), Maharashtra, Pune-30 and has been verified by them.
The Coordination Committee, established under Government Resolution by GR No. Abhyas - 2116/ (Pra.Kra.43/16) SD - 4 dated April 25, 2016, approved this textbook in its meeting held on April 16, 2025 for use from academic year 2025-26 onwords.

MATHEMATICS

STANDARD ONE

समग्र शिक्षा

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Scan the adjacent QR code with your smartphone to access digital resources, including audio, visual and audio-visual materials, that complement this textbook.

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The Constitution of India

Preamble

WE, THE PEOPLE OF INDIA, having solemnly resolved to constitute India into a SOVEREIGN SOCIALIST SECULAR DEMOCRATIC REPUBLIC and to secure to all its citizens:

JUSTICE, social, economic and political;

LIBERTY of thought, expression, belief, faith and worship;

EQUALITY of status and of opportunity; and to promote among them all

FRATERNITY assuring the dignity of the individual and the unity and integrity of the Nation;

IN OUR CONSTITUENT ASSEMBLY this twenty-sixth day of November, 1949, do HEREBY ADOPT, ENACT AND GIVE TO OURSELVES THIS CONSTITUTION.

The Constitution of India

Chapter IV A

Fundamental Duties

ARTICLE 51A

Fundamental Duties – It shall be the duty of every citizen of India –

(a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;

(b) to cherish and follow the noble ideals which inspired our national struggle for freedom;

(c) to uphold and protect the sovereignty, unity and integrity of India;

(d) to defend the country and render national service when called upon to do so;

(e) to promote harmony and the spirit of common brotherhood amongst all the people of India transcending religious, linguistic and regional or sectional diversities, to renounce practices derogatory to the dignity of women;

(f) to value and preserve the rich heritage of our composite culture;

(g) to protect and improve the natural environment including forests, lakes, rivers and wild life and to have compassion for living creatures;

(h) to develop the scientific temper, humanism and the spirit of inquiry and reform;

(i) to safeguard public property and to abjure violence;

(j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement;

(k) who is a parent or guardian to provide opportunities for education to his child or, as the case may be, ward between the age of six and fourteen years

NATIONAL ANTHEM

Jana-gana-mana-adhināyaka jaya hē
Bhārata-bhāgya-vidhātā,

Panjāba-Sindhu-Gujarāta-Marāthā
Drāvida-Utkala-Banga

Vindhya-Himāchala-Yamunā-Gangā
uchchala-jaladhi-taranga

Tava subha nāmē jāgē, tava subha āsisa māgē,
gāhē tava jaya-gāthā,

Jana-gana-mangala-dāyaka jaya hē
Bhārata-bhāgya-vidhātā,

Jaya hē, Jaya hē, Jaya hē,
Jaya jaya jaya, jaya hē.

PLEDGE

India is my country. All Indians are my brothers and sisters.

I love my country, and I am proud of its rich and varied heritage. I shall always strive to be worthy of it.

I shall give my parents, teachers and all elders respect, and treat everyone with courtesy.

To my country and my people, I pledge my devotion. In their well-being and prosperity alone lies my happiness.

राज्यगीत

जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ।।धृ।।

भिती न आम्हा तुझी मुळी ही गडगडणाऱ्या नभा
अस्मानाच्या सुलतानीला जबाब देती जिभा
सह्याद्रीचा सिंह गर्जतो शिव शंभू राजा
दरी दरीतून नाद गुंजला महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा ।।१।।

काळ्या छातीवरी कोरली, अभिमानाची लेणी
पोलादी मनगटे खेळती, खेळ जीवघेणी
दारिद्र्याच्या उन्हात शिजला
निढळाच्या घामाने भिजला
देश गौरवासाठी झिजला
दिल्लीचे ही तख्त राखितो, महाराष्ट्र माझा
जय जय महाराष्ट्र माझा, गर्जा महाराष्ट्र माझा
जय जय महाराष्ट्र माझा ।।२।।

PREFACE

Greeting to my small friends, teachers and parents,

It is great pleasure that, we present to you the Standard One Mathematics textbook prepared in accordance with the National Educational Policy 2020. According to the State Curriculum Framework 2024, it is suggested that easy and attractive books be considered for the age group of six to eight. The content in this textbook should not be just in the form of instructions but should provide opportunities for children to work on their own.

In this textbook, the emotional world of children and their preferences have been fully considered. Considering the interest of these children entering in standard one, this textbook has been provided with many colourful and attractive pictures. In order to enrich the content while thinking from a mathematical perspective, each chapter has been started by using various objects and situations from the emotional world of children as well as their surroundings. Concepts are explained in simple language. To give children the opportunity to practise, they are given sufficient examples to practise and enough space to solve those examples. For further reinforcement of the concepts suggestions have been given for teachers. An attempt has been made to make the book interesting through riddles, songs etc.

The illustrations and examples, provided in this text book are not only child-friendly but also socially balanced. Efforts have been taken to select the images, pictures with a view to ensure that all sections of society receive adequate priority. The Panchkosh and 21st century skills mentioned in the NEP 2020 have been incorporated into various components of the textbook. The content of the book is relevant to the development sector. An attempt has been made to maintain ascending order of difficulty level and variety in the examples given in this textbook is consistent with the educational goals, curriculum objectives, competencies and learning outcomes. The textbook includes activities, games, stories and songs to make learning enjoyable for children.

Best wishes to all for a bright future.

(Krishnakumar Patil)
Director
Maharashtra State Bureau of Textbook
Production and Curriculum Research, Pune.

Pune
Date : 30 April 2025
Indian Solar Year : 10, Vaishakh, 1947
Akshayya Tritiya

Learning Competencies and Learning Outcomes

Curricular Goal : Children develop mathematical understanding and abilities to recognize the world through quantities, shapes and measures.

Learning Competencies	Learning Outcomes
(1) Sorts objects into groups and subgroups based on more than one property.	Child sorts objects according to properties and describes the rules of sorting. (e.g. Groups of animals that live in the same environment.)
(2) Recognizes and extends the simple pattern of shapes and numbers in their surroundings.	Fills in the missing element of simple repeating patterns in different aspects with adults help. (eg. red-blue, red-blue, red-blue)
(3) Counts the numbers upto 99 (after and before) in the groups of 10s and 20s	Counts objects greater than 20 using number names upto 99 and counts the numbers according to the groups of 10s.
	Counts forward and backward from a specific number. (0 and 99)
	Recognizes the numbers/quantities in group of 2. (e.g. two groups of 10 makes 20)
(4) Arranges the numbers upto 99 in ascending and descending order.	Arranges objects in the same group according to their various properties. (e.g. size, measure, length, weight)
	Arranges the given number in ascending and descending order. (1 to 9)
(5) Recognises and uses numerals and writes number names upto 99 using place value concept.	Recognizes the symbol zero to represent absence of object/thing.
	Recognizes and writes numerals upto 20 and writes number names upto 10.
	Compares two numbers upto 20 and uses vocabulary like bigger than or smaller than.
(6) Using flexible strategies and derives combinations of composing (add together) and decomposing (take away from the set) performs addition and subtraction of two-digit numbers.	Uses real life situations and concrete objects to solve addition sums upto 18 using addition fact.
	Uses real life situations and concrete objects to construct and solve subtraction problems upto 9 using subtraction facts. (e.g. taking away seeds and beads from the given set.)
	Develops relationship between addition and subtraction of numbers.
	Recognizes the symbols +/- for addition/subtraction operations.
(7) Recognises multiplication as repeated addition and division as equal sharing.	Solves multiplication problems by grouping for small numbers.
	Recognizes the symbol (×) for multiplication.
	Uses trial and error method and by sharing into groups for solving division problems.
	Recognizes the symbol for division.
(8) Indentifies, creates and classifies geometric shapes with their observable properties as well as understands and explains the spatial relationship of objects.	Uses and develops vocabulary of spatial relationship. (e.g. up - down, inside - outside, near - far, before - after)
	Collects objects of various shapes and sizes from the surroundings. (e.g. pebbles, boxes, balls, cones, pipes)
	Sorts, classifies and describes the objects on the basis of shapes and other observable properties.
	Observes and describes the physical features of various shapes in their own language. (e g. a ball rolls, a box slides and has corners)
	Compares shapes based on various properties. (e.g. length, area, volume)
(9) Selects tools and units for simple measurements of the length, weight and volume of objects in their immediate environment.	Distinguishes between near-far, thin-thick, longer/taller-shorter.
	Measures small lengths using different non standard units. (in the context of games.e.g. *Gilli Danda*, Marbles.)
	Estimates short distance and length and verifies using non-standard units. (e.g. figures, hand span, forearm, foot steps.)

	Compares and places in the order from light to heavy objects or vice-versa.
	Estimates and measures volumes of containers using uniform non-standard units like a cup/spoon/mug.
(10) Performs simple measurements of time in units of minutes, hours, days, weeks and months.	Differentiates between events that occur in time using terms like earlier and later.
	Take the qualitative feel of long and short duration of school days and holidays.
	Narrates the sequence of events in a day. Arranges the events/incidents of the day in order.
(11) Performs simple transactions using money upto Rupees 100.	Using coins and currency notes creates amounts upto Rupees 20.
(12) Developes sufficient and appropriate vocabulary to express understanding of related processes to quantities, shapes, space and measurement.	Constructs full sentences to describe a mathematical problem related to quantities, shapes, space and measurement.
(13) Formulates and solves simple mathematical problems related to quantities, shapes, space and measurement.	Recognizes situations in the real world as simple mathematical problems.
	Solves simple numerical problems using different strategies.

While handling the textbook....

Teachers parents and facilitators should....

- Explain mathematical concepts using songs, rhymes, games, stories, picture observations activities, projects, demonstrations, picture cards, number cards, beads / manimalas, material from the mathematics box and material available in the surroundings.
- Ask them to do activities individually, in pairs or in groups as per need. This will help to reinforce the concepts.
- Understanding and applying mathematical concepts as well as a journey from hads on to minds on should be made.
- It has been considered at some places in the textbook that students should be given higher opportunities such as constructing examples/questions themselves, rather than just solving examples/questions. It is to be expanded more and more.
- Engage students in the teaching learning process by asking students thought provoking questions and activities.
- An attempt has been made to include most of the learning outcomes in all the units/chapters.
- The instructions in the textbook are given as a guideline. Teacher should modify teaching and learning process according to local context, local conditions, dialect and examples from the surrounding.

Introduction to emojis used in the textbook

Converse.

Discuss.

Write.

Sing a song.

Classify.

Do activity.

Observe.

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